# THE FITTING-TYPE IDEAL ATTACHED TO A BOUNDED COMPLEX OF FINITE FREE MODULES IS WELL DEFINED UP TO POWERS OF THE NONZERODIVISOR DEPENDING ON THE EULER CHARACTERISTIC

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

*Editorial note.* The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

*Editorial note.* Two internal cross-references appear inside the counted proof span, \ref{lemma-compare-representatives-perfect} at line 27910 and \ref{lemma-ideals-generated-by-minors} at line 27960. Both point to lemmas of this same chapter that lie outside the counted span; they are recorded as navigation pointers only, and both applications together with the surrounding computation stay inside the counted proof span.

*Allowed standard prerequisites (editorial).*

- Ideals I_i of minors of an augmented map and the behaviour of these ideals for block matrices. invoked through the cited Lemma lemma-ideals-generated-by-minors at proof line 27960 and applied at proof lines 27961-27963. Gives the formula $\mathrm{I_\{r_i+1\}(T)} = \mathrm{I_\{r_i+1\}(T_1)} + \mathrm{I_\{r_i\}(T_1)}$ that collapses the two ideals in the case j = i.
- Localization at primes to test an equality of ideals. used at proof lines 27908-27909. Reduces the general statement to the local case in which the comparison with a trivial complex can be made.
- The trivial two-term complex concentrated in two adjacent degrees with identity differential. defined at proof lines 27913-27917. Is the summand Q^bullet whose addition changes I_i only by a power of f, which is exactly what the exponents m and n record.

*Source reference.* Original lemma 1: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19628–19628. Same-chapter pointer invoked at proof line 27910 to reduce two complexes representing the same derived object to the case where one is a summand of the other; the reduction, the induction the author states is omitted, and every subsequent case distinction stay inside the counted proof span.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 2: sources/stacks-more-algebra/pinned-original-source.tex, source lines 1326–1326. Same-chapter pointer supplying the standard formula for the ideals generated by the minors of a block matrix, used at proof line 27960 to collapse I_{r_i+1}(T) into I_{r_i}(T_1); the computation that consumes it, including the values I_0(T_2) = A, I_1(T_2) = A and I_p(T_2) = 0 for p >= 2, stays inside the counted proof span.

**Lemma 0.2.** *Let $A$ be a ring and let $f \in A$ be a nonzerodivisor. Let $M^\bullet$ and $N^\bullet$ be two bounded complexes of finite free $A$-modules representing the same object of $D(A)$. Then*

$$f^m I_i(M^\bullet, f) = f^n I_i(N^\bullet, f)$$

*as ideals of $A$ for integers $n, m \geq 0$ such that*

$$m + \sum\nolimits_{j \geq i} (-1)^{j-i} rk(M^j) = n + \sum\nolimits_{j \geq i} (-1)^{j-i} rk(N^j)$$

*Proof.* It suffices to prove the equality after localization at every prime ideal of $A$. Thus by Lemma 1 and an induction argument we omit we may assume $N^\bullet = M^\bullet \oplus Q^\bullet$ for some trivial complex $Q^\bullet$, i.e.,

$$Q^\bullet = \ldots \to 0 \to A \xrightarrow{1} A \to 0 \to \ldots$$

where $A$ is placed in degree $j$ and $j+1$. If $j \neq i-1, i, i+1$ then we clearly have equality $I_i(M^\bullet, f) = I_i(N^\bullet, f)$ and $m = n$ and we have the desired equality. If $j = i+1$ then the maps

$$(f, d^i) : M^i \to M^i \oplus M^{i+1} \quad \text{and} \quad (f, d^i, 0) : M^i \to M^i \oplus M^{i+1} \oplus A$$

have the same nonzero minors hence in this case we also have $I_i(M^\bullet, f) = I_i(N^\bullet, f)$ and $m = n$. If $j = i$, then $I_i(M^\bullet, f)$ is the ideal generated by the $r_i \times r_i$-minors of

$$(f, d^i) : M^i \to M^i \oplus M^{i+1}$$

and $I_i(N^\bullet, f)$ is the ideal generated by the $(r_i + 1) \times (r_i + 1)$-minors of

$$(f \oplus f, d^i \oplus 1) : (M^i \oplus A) \to (M^i \oplus A) \oplus (M^{i+1} \oplus A)$$

With suitable choice of coordinates we see that the matrix of the second map is in block form

$$T = \begin{pmatrix} T_1 & 0 \\ 0 & T_2 \end{pmatrix}, \quad T_1 = \text{matrix of first map}, \quad T_2 = \begin{pmatrix} f \\ 1 \end{pmatrix}$$

With notation as in Lemma 2 we have $I_0(T_2) = A$, $I_1(T_2) = A$, $I_p(T_2) = 0$ for $p \geq 2$ and hence $I_{r_i+1}(T) = I_{r_i+1}(T_1) + I_{r_i}(T_1) = I_{r_i}(T_1)$ which means that $I_i(M^\bullet, f) = I_i(N^\bullet, f)$. We also have $m = n$ so this finishes the case $j = i$. Finally, say $j = i-1$. Then we see that $m = n + 1$, thus we have to show that $f I_i(M^\bullet, f) = I_i(N^\bullet, f)$. In this case $I_i(M^\bullet, f)$ is the ideal generated by the $r_i \times r_i$-minors of

$$(f, d^i) : M^i \to M^i \oplus M^{i+1}$$

and $I_i(N^\bullet, f)$ is the ideal generated by the $(r_i + 1) \times (r_i + 1)$-minors of

$$(f \oplus f, d^i) : (M^i \oplus A) \to (M^i \oplus A) \oplus M^{i+1}$$

With suitable choice of coordinates we see that the matrix of the second map is in block form

$$T = \begin{pmatrix} T_1 & 0 \\ 0 & T_2 \end{pmatrix}, \quad T_1 = \text{matrix of first map}, \quad T_2 = \begin{pmatrix} f \end{pmatrix}$$

Arguing as above we find that indeed $f I_i(M^\bullet, f) = I_i(N^\bullet, f)$. □